

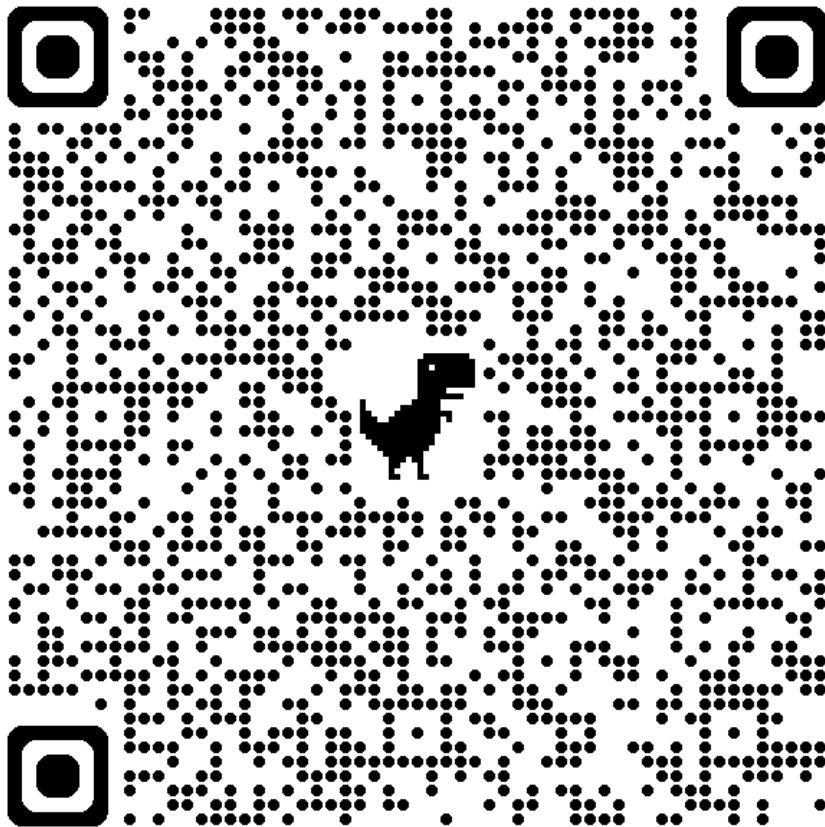


Outline:

- Atomic Densities
 - Linear Density
 - Planar Density
- Single- vs poly- crystalline materials
- X-ray Diffraction
- Example
- Polymorphism and Allotropy

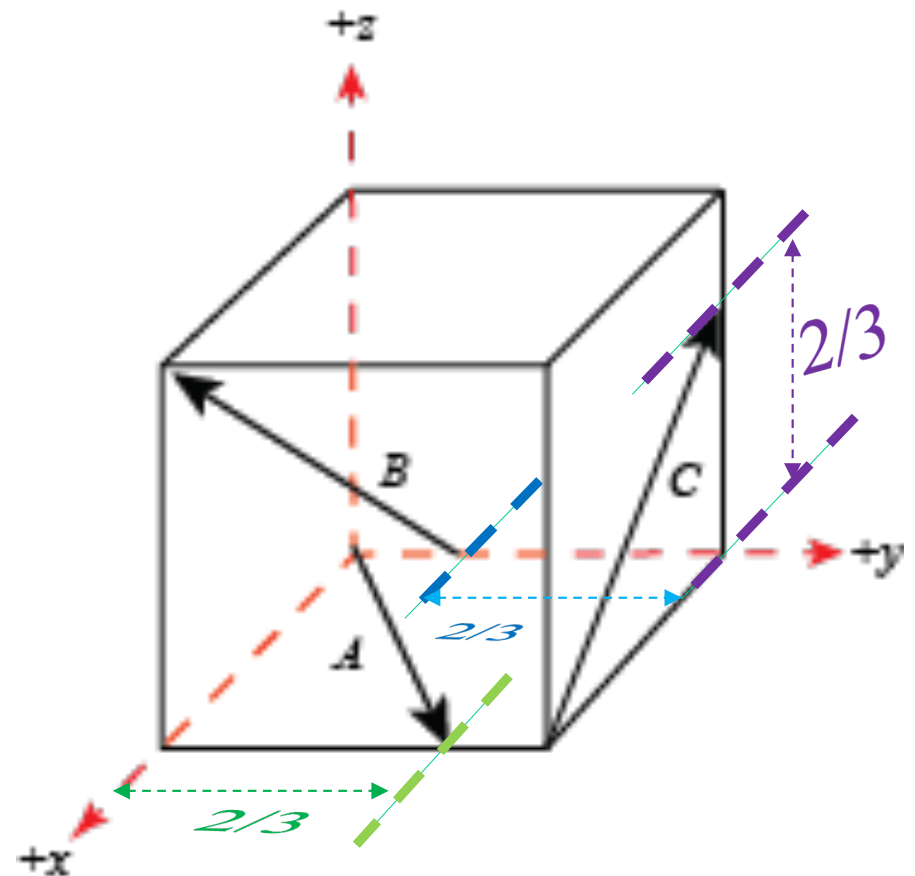


Review - Directions



Use the QR code shown in the classroom. This one won't work.

Which direction is $[\bar{3}\bar{1}3]$





Atomic Densities

- **Linear Density**

- Number of atoms per length whose centers lie on the direction vector for a specific crystallographic direction.

$$\text{Linear Density} = \frac{\text{Number of atoms centered on direction vector}}{\text{length of direction vector}}$$

- **Planar Density**

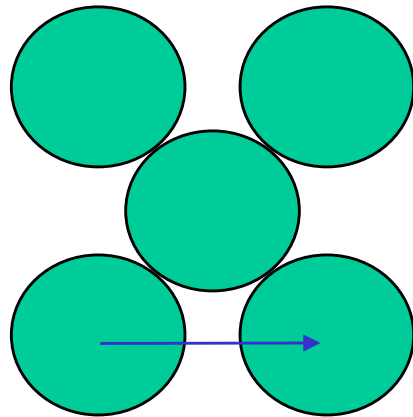
- Number of atoms per unit area that are centered on a particular crystallographic plane.

$$\text{Planar Density} = \frac{\text{Number of atoms centered on a plane}}{\text{area of the plane}}$$



Linear Density

- Calculate the linear density of the [100] direction for the FCC crystal



$$L_D = n/L_L \quad \text{linear density}$$

$$n = 1 \quad \text{atoms}$$

$$L_L = a \quad \text{line length}$$

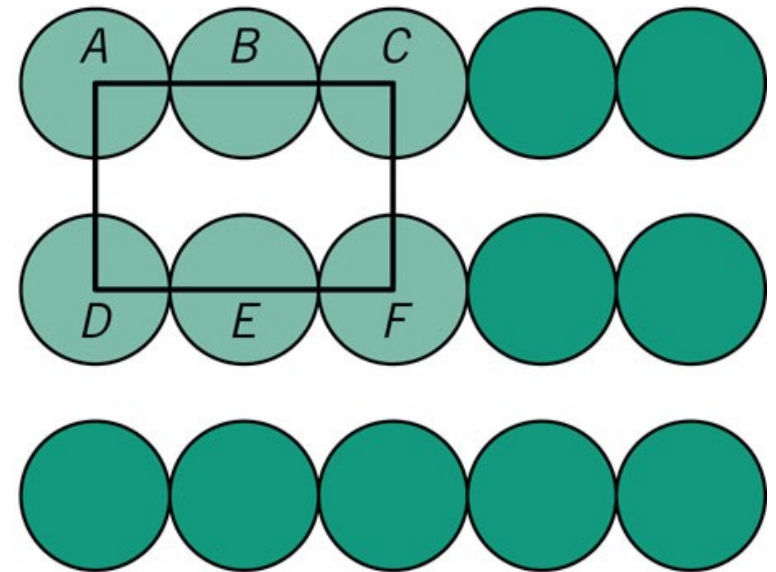
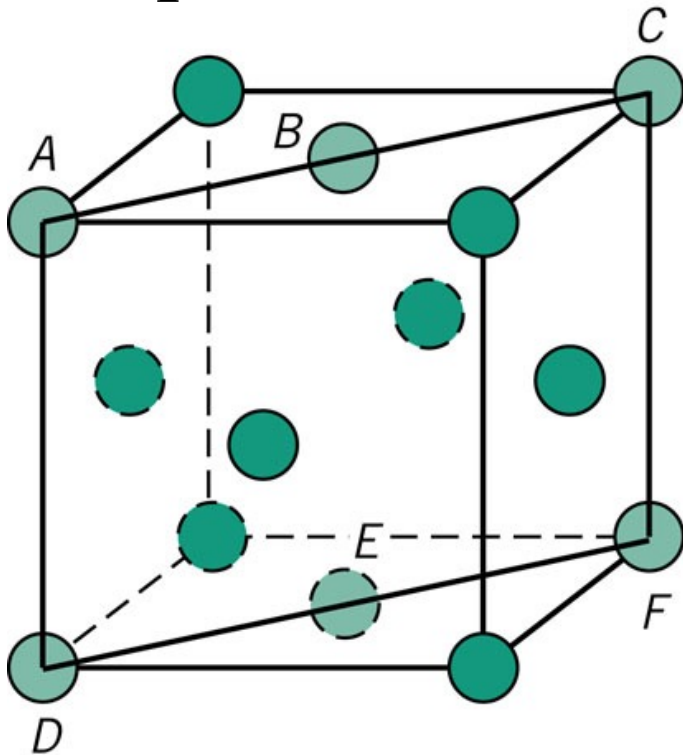
$$\text{For FCC } a = \dots\dots\dots$$

$$L_D = \dots\dots\dots$$



Planar Density

- Calculate the planar density of the (110) plane for the FCC crystal

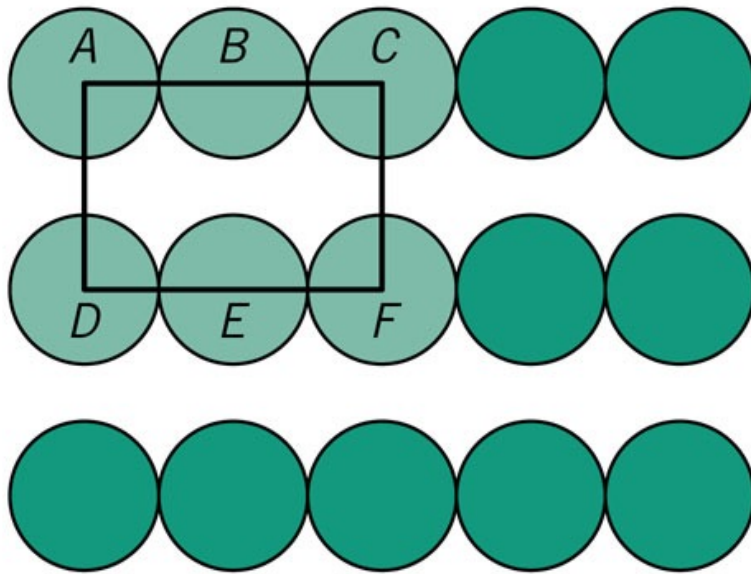


- Compute planar area
- Compute number of atoms

For an atom to be counted, it has to be centered on that plane.



Planar Density



Plane area $A_p = (AC) \times (AD)$

$$AC = 4R$$

$$AD = 2R\sqrt{2}$$

$$A_p = (4R)(2R\sqrt{2}) = 8R^2\sqrt{2}$$

Number of atoms =

$$PD = \frac{n}{A_p} = \underline{\hspace{2cm}}$$



Linear and Planar Density

- Why do we care?
- Properties, in general, depend on linear and planar density.
- Examples:
 - Speed of sound along directions
 - Slip (..... *in metals*) depends on linear and planar density
 - Slip occurs on planes that have the greatest density of atoms in direction with highest density

we would say along closest packed directions on the closest packed planes



Crystals As Building Blocks

- *Some* engineering applications require **single** crystals:



Single crystal

(Courtesy P.M. Anderson)

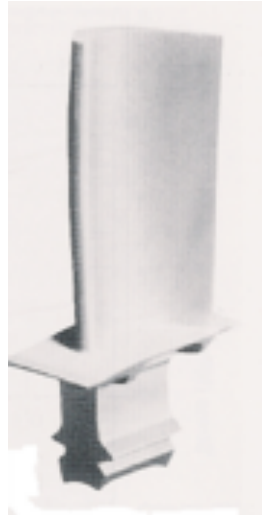


Fig. 8.30(c), *Callister 6e*.
(courtesy of Pratt and Whitney)

- Turbine blades
- Nickel alloy – single crystal
- to improve high temp. mechanical properties



(Courtesy GE Superabrasives)

diamond single crystals
for abrasives

- *Most* engineering materials are

- Each "grain" is a **single** crystal.
- If crystals are **randomly** oriented, overall component properties are **not** directional.
- Crystal sizes typically range from 1 nm to 2 cm (i.e., *from a few to millions of atomic layers*).



Nb-Hf-W plate with an electron beam weld

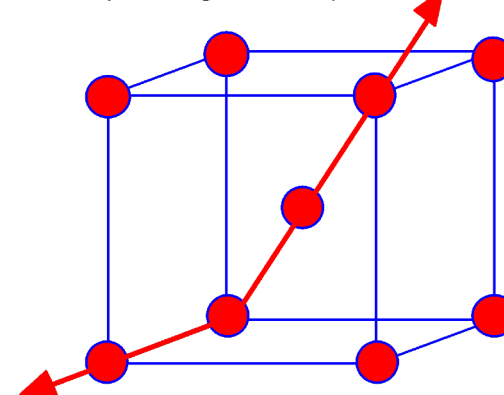


Single Vs Polycrystals

- **Single** Crystals

- Properties vary with direction:
- Example: the modulus of elasticity (E) in BCC iron:

E (diagonal) = 273 GPa

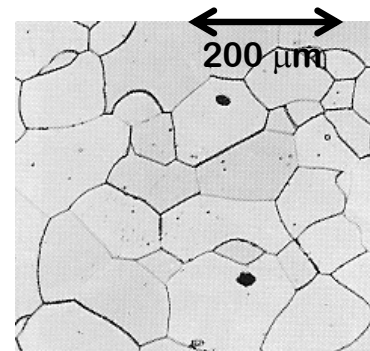


Data from Table 3.3,
Callister 6e.

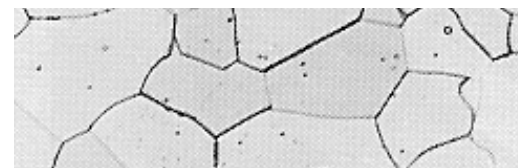
E (edge) = 125 GPa

- **Polycrystals**

- Properties may/may not vary with direction.
- If grains are randomly oriented:
($E_{poly iron} = 210 \text{ GPa}$)
- If grains are **textured**, anisotropic.



Adapted from Fig.
4.12(b), *Callister 6e.*

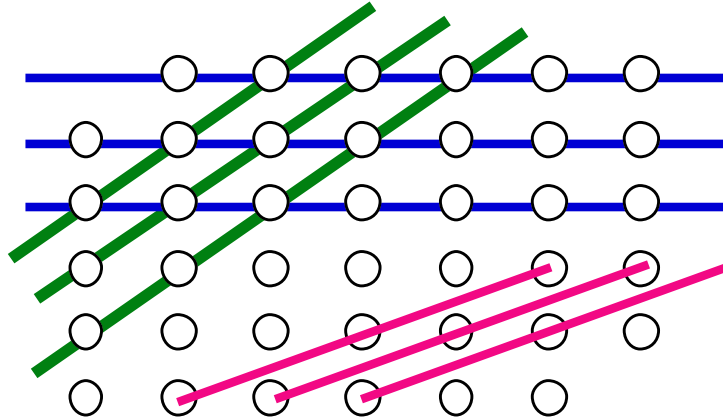


Textured grains



Inter-Planar Spacing & X-Ray Diffraction

2D



- Inter-planar spacing
 - The inter-planar spacing in a **particular direction** is the distance between equivalent planes of atoms
- The existence of planes and the distances between different sets of them in a material are characteristic for each material
 - ➔ Hence, inter-planar spacings are measured by **x-ray diffraction** to identify unknown materials!

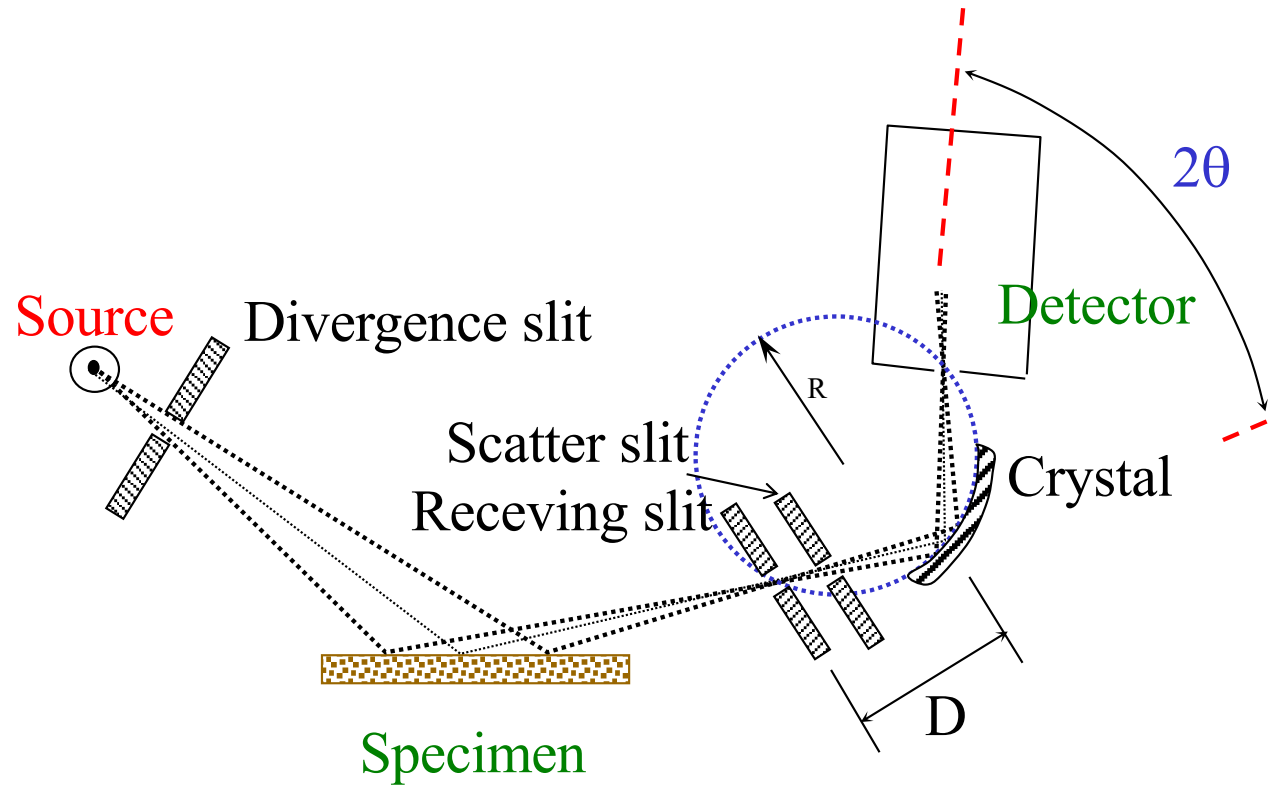


X-Ray Diffraction

- Can be used to determine **crystal structure**
 - identify **unknown** materials
 - measure lattice parameters
- X-rays are a form of electromagnetic radiation that have high energies and short wavelengths.
- Diffraction occurs whenever a wave encounters a series of regularly spaced obstacles that;
 - Can scatter the wave
 - Have a spacing comparable to the
- **X-ray wavelength** (λ) $\sim$ inter-atomic spacing.
- Other techniques such as **neutron** or **electron** diffraction, also, can be used.

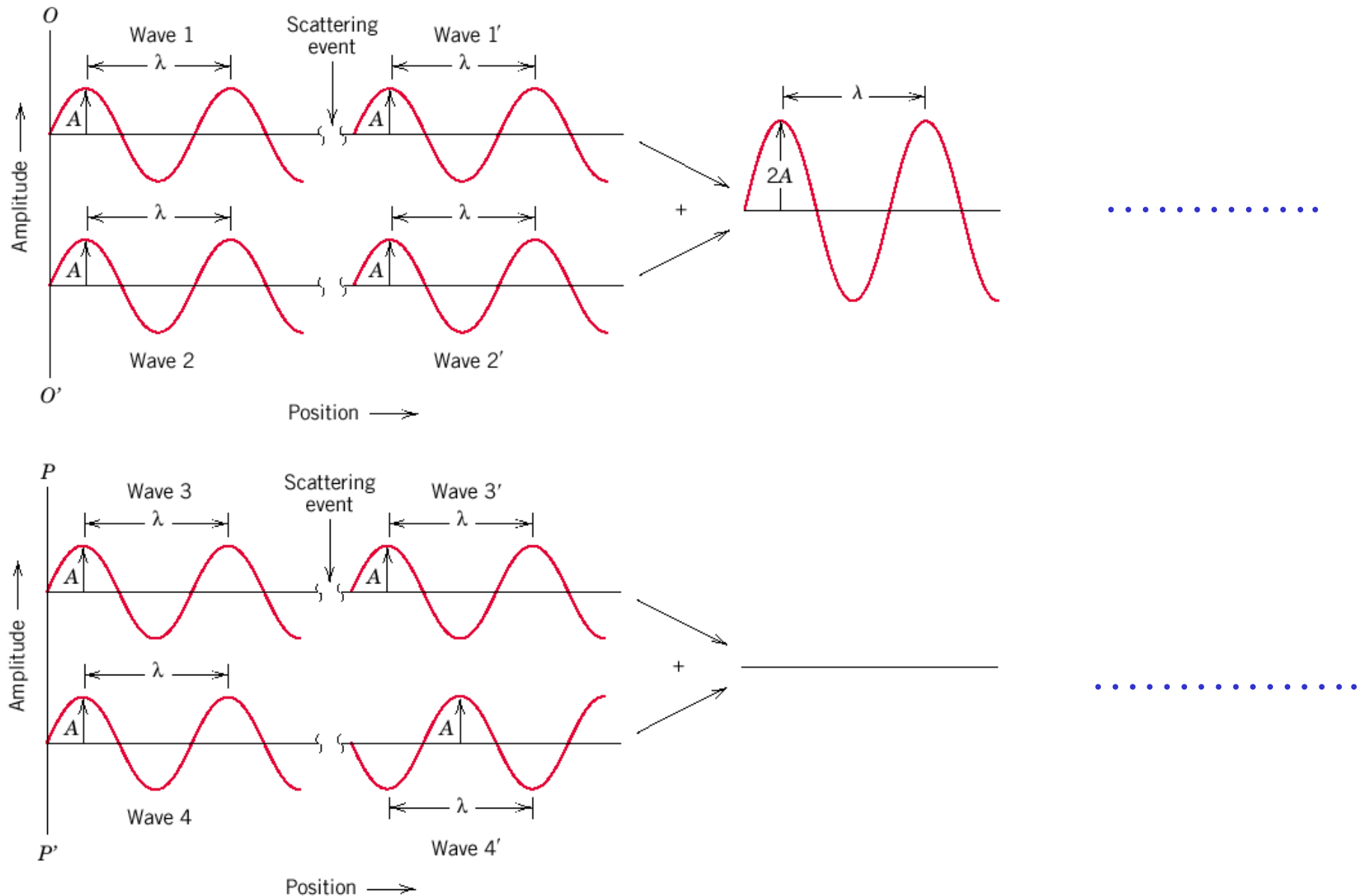


X-ray Diffractometer



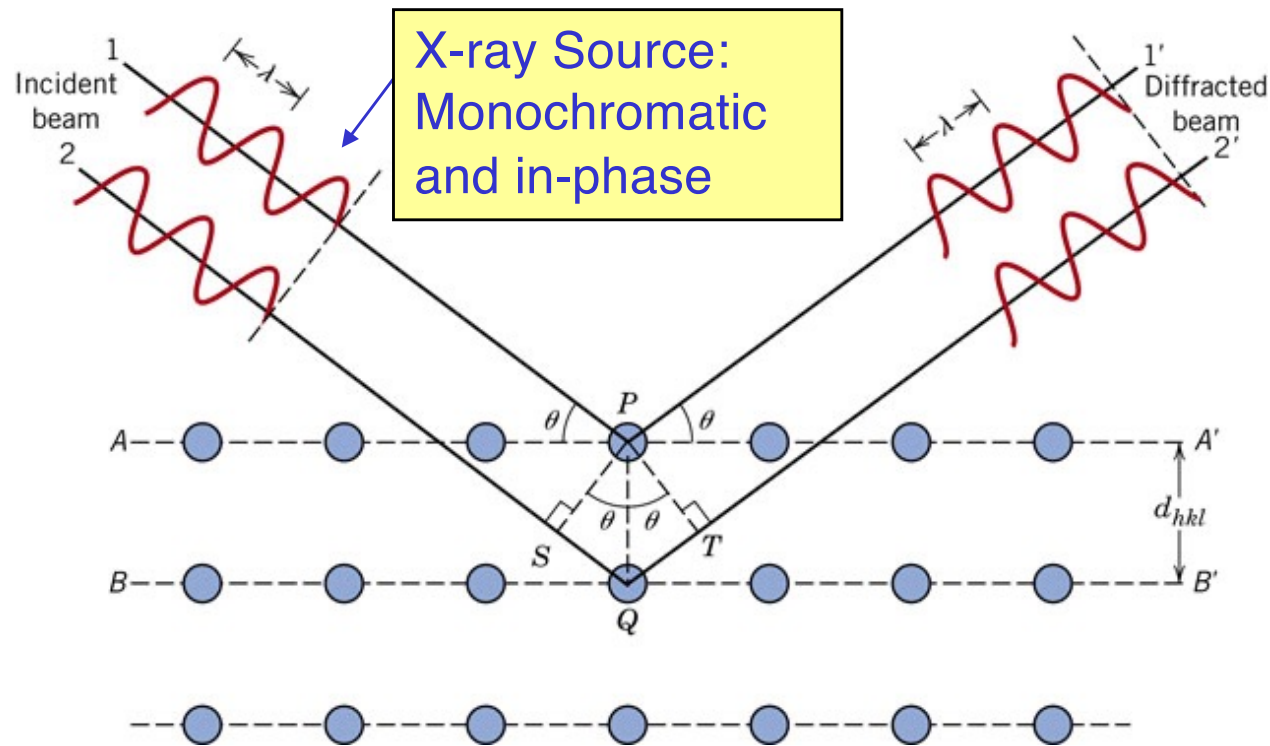


Constructive & Destructive Interference





Bragg's Law



The law: For constructive interference, the additional path length $SQ + QT$ must be an integer number of wavelength (λ).

$$n\lambda = SQ + QT = d_{hkl}\sin\theta + d_{hkl}\sin\theta = 2 d_{hkl}\sin\theta$$

$n = 1, 2, 3 \dots$ order of reflection



Bragg's Law

$$n\lambda = d_{hkl} \sin \theta + d_{hkl} \sin \theta$$

$$= 2d_{hkl} \sin \theta$$

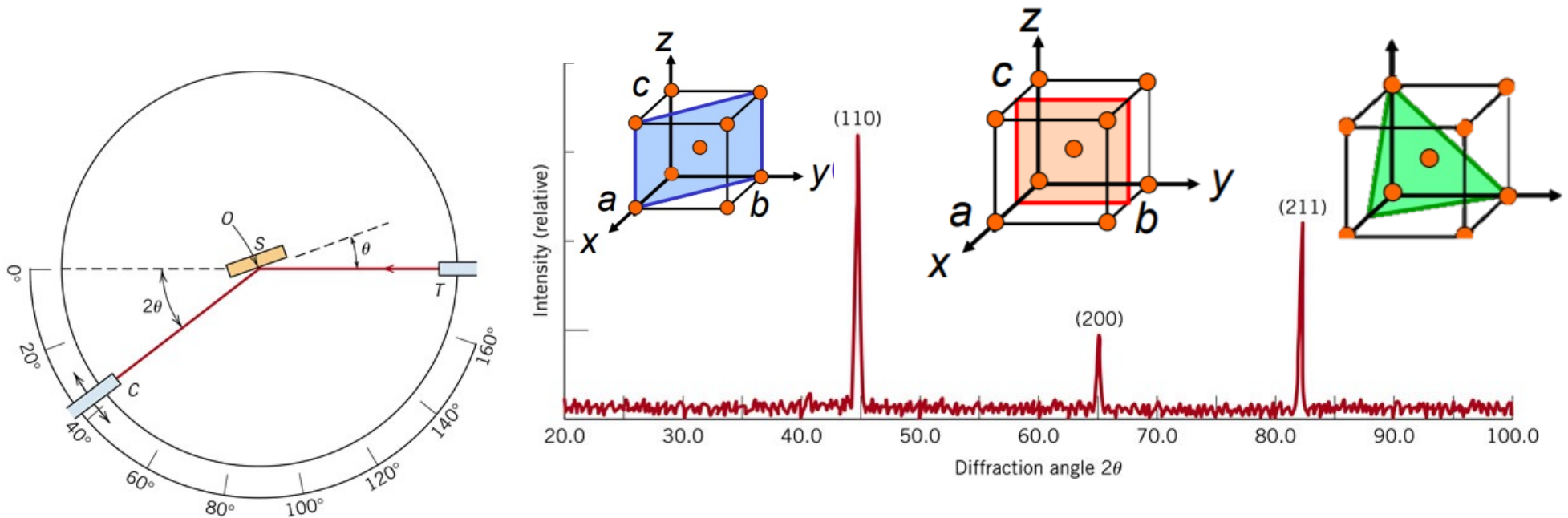
- We have a simple expression relating the **x-ray wavelength** and **interatomic spacing** to the angle of the diffracted beam.
- If **Bragg's law** is not satisfied, then the interference will be **nonconstructive** in nature so as to yield a very low-intensity diffracted beam.
- Magnitude of difference between two adjacent and parallel planes of atoms is function of **Miller Indices** and the **lattice parameter**. For cubic symmetry:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$



Diffractometer Technique

- Use **powder** (or polycrystalline) sample to guarantee some particles will be oriented properly such that every possible set of crystallographic planes will be available for diffraction.



Each material has a unique set of planar distances and extinctions, making X-ray diffraction useful in analysis of an unknown material.



Example

For BCC iron (Fe), compute:

- (a) the **interplanar spacing** of the (220) planes, and
- (b) the **diffraction angle** at which the (220) reflection occurs.
 - Given: *lattice parameter of Fe, $a = 0.2866 \text{ nm}$* ;
 - *X-ray wavelength, $\lambda = 0.1790 \text{ nm}$* ;
 - *first-order reflection ($n = 1$)*.

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\sin \theta = \frac{n\lambda}{2d_{hkl}}$$

$$\theta = \dots\dots\dots^\circ$$

$$2\theta = \dots\dots\dots^\circ$$



Polymorphism and Allotropy

- Some materials may have more than one **crystal structure** depending on **temperature** and **pressure** - called *POLYMORPHISM*
- Carbon
 - graphite
 - diamond
- Iron BCC and FCC



Next topic:
Imperfections in Solids